KS3 DIGITAL TECHNOLOGY

ALGORITHMS, FEEDS AND DIGITAL WELLBEING

What decides what you see online: how recommendation algorithms rank your feed, how they learn from your behaviour, how filter bubbles and rabbit holes form, how persuasive design competes for your attention, how to take control of your digital wellbeing, and how to design a fairer feed.

MR BUTLER'S DIGITAL TECHNOLOGY

LESSONS

1 Who chooses your feed?

2 How recommenders learn

3 Bubbles and rabbit holes

4 The attention economy

5 Your digital wellbeing

6 Designing fairer feeds

LESSON 1 OF 6

WHO CHOOSES YOUR FEED?

A recommendation algorithm chooses and orders content for each user.

STARTER · 5 MINUTES

DO NOW

Open any video app in your head. Why do you think the first video it shows you is that one?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain what a recommendation algorithm is, compare chronological and ranked feeds, and describe the signals algorithms use

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what a recommendation algorithm does
- ✓ I can compare chronological and ranked feeds
- ✓ I can describe the signals algorithms use

THE ALGORITHM

- An algorithm is a set of steps to solve a problem
- A recommendation algorithm chooses what to show each person
- Huge scale: billions of posts, billions of users, decisions in milliseconds
- Used for videos, music, shopping, news, games and adverts

KEY WORDS

ALGORITHM	A precise set of steps to solve a problem.
RECOMMENDATION ALGORITHM	An algorithm that chooses and orders content for each user.
RANKED FEED	A feed ordered by an algorithm's prediction of what you'll engage with.
CHRONOLOGICAL FEED	A feed that shows posts in time order, newest first.
ENGAGEMENT	How much people interact with content, such as watching, liking and sharing.

TWO KINDS OF FEED

- Chronological feed: newest posts first, from accounts you follow
- Ranked feed: an algorithm orders posts by what it predicts you'll engage with
- "For You" feeds show content from accounts you don't follow
- Most big platforms default to ranked feeds

ACTIVITY · 20 MINUTES · PAIRS

FEED DETECTIVE

1. Look at the two (fictional) feeds for "Cara": one chronological, one ranked.
2. Describe three differences.
3. For five posts in the ranked feed, suggest which signals made the algorithm show them.
4. Discuss: which feed would you prefer, and why?

In your workbook, answer:

? Describe three differences between the two feeds.

? Which signals might explain a football video appearing in Cara's ranked feed?

Your workbook: Lesson 1 · tick each step off as you go

WHAT THE ALGORITHM WATCHES

- Watch time: how long you watch, and whether you rewatch
- Likes, comments, shares, saves, follows
- What you skip or scroll past quickly
- Your device, location, time of day, language, and what similar users like

QUICK CHECK

1. What is a recommendation algorithm?

A A list of passwords
B A search engine only
C A type of virus
D An algorithm that chooses what content to show each person

2. How is a chronological feed ordered?

A Randomly
B By predicted interest
C Newest posts first
D Alphabetically

3. Which is a signal algorithms use?

A Your shoe size
B Watch time
C The weather on Mars
D Your exam results

4. What does engagement mean?

A A password
B A type of feed
C How much people interact with content
D Getting married

QUICK CHECK: ANSWERS

1. **D: An algorithm that chooses what content to show each person**
   It personalises feeds.

2. **C: Newest posts first**
   Time order.

3. **B: Watch time**
   How long you watch.

4. **C: How much people interact with content**
   Likes, shares, watch time.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between a chronological feed and a ranked feed.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what a recommendation algorithm does
- ✓ I can compare chronological and ranked feeds
- ✓ I can describe the signals algorithms use

LESSON 2 OF 6

HOW RECOMMENDERS LEARN

Recommenders build a profile of each user from their behaviour.

STARTER · 5 MINUTES

DO NOW

If you told a friend your three favourite songs, how would they choose a fourth to recommend?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how recommender systems use data, collaborative filtering and content-based filtering to make predictions

SUCCESS CRITERIA: I CAN...

- ✓ I can explain content-based filtering
- ✓ I can explain collaborative filtering
- ✓ I can describe how machine learning scores and ranks content

TWO BIG IDEAS

- Content-based: "You liked this, so here's something similar"
- Collaborative: "People like you liked this"
- Most systems combine both
- Machine learning predicts a score for every possible item

KEY WORDS

USER PROFILE	A record of a person's behaviour and information used to personalise content.
CONTENT-BASED FILTERING	Recommending items similar to ones a user has liked.
COLLABORATIVE FILTERING	Recommending items that similar users have liked.
COLD START	The problem of recommending to new users, or recommending new items, with no history.
EXPLORATION	When a recommender shows something new to test a user's interest.

SCORING AND RANKING

- Step 1: find thousands of candidates
- Step 2: predict how likely you are to engage with each one
- Step 3: rank by score, mix in variety, remove anything against the rules
- Step 4: learn from what you do next

ACTIVITY · 25 MINUTES · GROUPS OF 5

HUMAN RECOMMENDER

1. Each person rates ten items (e.g. games, songs or snacks from the card list) from 1 to 5, leaving two blank.
2. Use collaborative filtering: find who in your group is most similar to you.
3. Predict your two blank ratings using that person's ratings.
4. Check how accurate the predictions were, then discuss when the method worked and when it failed.

In your workbook, answer:

? Who was most similar to you, and how did you decide?

? When did the method fail, and why?

Your workbook: Lesson 2 · tick each step off as you go

PROBLEMS

- Cold start: new users or new posts have no data
- Exploration: algorithms sometimes test new things on you
- Popularity bias: popular content gets shown more and more
- Feedback loops: what you're shown shapes what you click

QUICK CHECK

1. What is content-based filtering?

A Recommending items similar to ones you liked
B Filtering coffee
C Recommending what similar users liked
D Blocking all content

2. What is collaborative filtering?

A Recommending only new items
B Recommending what similar users liked
C Working in a group
D Removing comments

3. What is the cold start problem?

A A frozen phone
B A slow computer in winter
C New users or posts have no data to base recommendations on
D A type of virus

4. Why do recommenders sometimes show you something new?

A By accident only
B They never do
C To explore and test your interests
D To annoy you

QUICK CHECK: ANSWERS

1. **A: Recommending items similar to ones you liked**
   Similar items.

2. **B: Recommending what similar users liked**
   People like you.

3. **C: New users or posts have no data to base recommendations on**
   No history yet.

4. **C: To explore and test your interests**
   Exploration.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how collaborative filtering could recommend a song.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain content-based filtering
- ✓ I can explain collaborative filtering
- ✓ I can describe how machine learning scores and ranks content

LESSON 3 OF 6

BUBBLES AND RABBIT HOLES

Personalised feeds can create filter bubbles, where you mostly see content that matches your existing interests and views.

STARTER · 5 MINUTES

DO NOW

If you only ever saw posts from people who agreed with you, how would your view of the world change?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain filter bubbles, echo chambers and rabbit holes, and evaluate their effects on individuals and society

SUCCESS CRITERIA: I CAN...

- ✓ I can explain filter bubbles and echo chambers
- ✓ I can explain how rabbit holes can lead to more extreme content
- ✓ I can evaluate the evidence and suggest ways to break out

FILTER BUBBLES

- Personalisation means everyone sees a different internet
- A filter bubble: you mostly see content matching your interests and views
- You may not realise what you're missing
- Search results, news feeds and videos can all be personalised

KEY WORDS

FILTER BUBBLE	When algorithms mostly show you content matching your existing interests and views.
ECHO CHAMBER	An online space where people mostly hear opinions that match their own.
RABBIT HOLE	When recommendations lead step by step to more extreme content.
POLARISATION	Groups becoming more extreme and further apart in their views.
CONFIRMATION BIAS	The tendency to prefer information that agrees with what we already believe.

ECHO CHAMBERS AND RABBIT HOLES

- Echo chamber: a group where everyone repeats the same views
- Confirmation bias: we prefer information that agrees with us
- Rabbit hole: each recommendation slightly more extreme than the last
- Topics: diet and fitness, conspiracy theories, misogyny, violence

WHAT'S THE EVIDENCE?

- Researchers disagree about how strong bubbles are
- Many people still see a mix of views
- But some users, especially young people, are shown harmful content quickly
- Ofcom's rules now require platforms to tame algorithms for children

ACTIVITY · 25 MINUTES · GROUPS OF 4

BUBBLE SIMULATION

1. Each pupil gets a different "starting interest" card and follows the recommendation flowchart for 8 rounds, picking the most "engaging" option each time.
2. Record the path you took and where you ended up.
3. Compare paths: how different were your final feeds? Did any become extreme?
4. Redesign one step of the flowchart to stop the rabbit hole.

In your workbook, answer:

? Describe your path and where it ended.

? How would you redesign the algorithm to stop the rabbit hole?

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What is a filter bubble?

A Mostly seeing content that matches your existing interests and views

B A soap bubble

C A type of feed setting

D A spam filter

2. What is an echo chamber?

A A room with an echo

B A music app

C A type of filter

D A space where people mostly hear views they already agree with

3. What is confirmation bias?

A Confirming your email

B Preferring information that agrees with what we already believe

C A type of algorithm

D A fact-check

4. What is a rabbit hole online?

A A broken link

B A game

C Recommendations leading step by step to more extreme content

D A pet video

QUICK CHECK: ANSWERS

1. **A: Mostly seeing content that matches your existing interests and views**
   Personalisation's side effect.

2. **D: A space where people mostly hear views they already agree with**
   Repeated views.

3. **B: Preferring information that agrees with what we already believe**
   Human tendency.

4. **C: Recommendations leading step by step to more extreme content**
   Gradual escalation.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how a filter bubble could form.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain filter bubbles and echo chambers
- ✓ I can explain how rabbit holes can lead to more extreme content
- ✓ I can evaluate the evidence and suggest ways to break out

LESSON 4 OF 6

THE ATTENTION ECONOMY

Many free apps make money from advertising, so they earn more when people spend more time on them.

STARTER · 5 MINUTES

DO NOW

Why are most social media apps free? How do they make money?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain the attention economy and identify persuasive design techniques used to keep people engaged

SUCCESS CRITERIA: I CAN...

- ✓ I can explain the attention economy
- ✓ I can identify persuasive design techniques
- ✓ I can explain variable rewards and why they're powerful

YOU ARE THE PRODUCT?

- Most social apps are free to use
- They make money from adverts
- More time on the app = more adverts = more money
- Your attention is what's being sold: the attention economy

KEY WORDS

ATTENTION ECONOMY	Companies competing for people's time and attention, often to sell advertising.
PERSUASIVE DESIGN	Design that uses psychology to influence what people do.
INFINITE SCROLL	A design where new content keeps loading so there's no end to the page.
VARIABLE REWARD	A reward that comes unpredictably, which makes people keep checking.
STREAK	A count of how many days in a row someone has used an app or done something.

PERSUASIVE DESIGN

- Infinite scroll: no natural stopping point
- Autoplay: the next video starts automatically
- Notifications: pulling you back in
- Streaks and badges: fear of losing progress

VARIABLE REWARDS

- Pull-to-refresh is like a slot machine
- Sometimes great content, sometimes nothing
- Unpredictable rewards are especially hard to stop checking
- Likes arrive at unpredictable times too

ACTIVITY · 20 MINUTES · PAIRS

DESIGN DETECTIVES

1. Look at screenshots of four (fictional) apps.
2. Spot and label every persuasive design technique you can find.
3. For each technique, decide: does it mostly help the user or the company?
4. Suggest a "humane" alternative for the two most problematic techniques.

In your workbook, answer:

? Which persuasive techniques did you find, and who do they mostly help?

? Suggest humane alternatives for two techniques.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is the attention economy?

A A bank
B Companies competing for people's time and attention to sell advertising
C Paying attention in class
D Saving money

2. What is infinite scroll?

A A bug
B A very long scroll of paper
C A type of algorithm
D Content that keeps loading so there's no end

3. What is a variable reward?

A A reward that comes unpredictably
B A fixed prize
C A discount
D A salary

4. How do most free social media apps make money?

A Advertising
B Selling phones
C Charging for every post
D Government grants

QUICK CHECK: ANSWERS

1. **B: Companies competing for people's time and attention to sell advertising**
   Attention is the product.

2. **D: Content that keeps loading so there's no end**
   No stopping point.

3. **A: A reward that comes unpredictably**
   Like a slot machine.

4. **A: Advertising**
   Ads pay for "free".

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why free apps want users to spend as much time as possible on them.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain the attention economy
- ✓ I can identify persuasive design techniques
- ✓ I can explain variable rewards and why they're powerful

LESSON 5 OF 6

YOUR DIGITAL WELLBEING

Technology can support wellbeing through connection, creativity and learning, but some patterns of use can affect sleep, mood, concentration and self-esteem.

STARTER · 5 MINUTES

DO NOW

How do you feel after an hour of scrolling compared with an hour of gaming with friends or making something?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how technology use can affect wellbeing and apply strategies and tools to stay in control

SUCCESS CRITERIA: I CAN...

- ✓ I can describe positive and negative effects of technology on wellbeing
- ✓ I can explain why what and how you use tech matters more than total time
- ✓ I can use tools and habits to stay in control of my feeds

THE GOOD AND THE NOT-SO-GOOD

- Positive: connection, creativity, learning, support communities, fun
- Not so good: poor sleep, comparison, distraction, feeling left out
- Effects differ from person to person
- How you feel afterwards is a useful guide

KEY WORDS

DIGITAL WELLBEING	Using technology in a way that supports physical and mental health.
SCREEN TIME	The amount of time spent using devices with screens.
SOCIAL COMPARISON	Judging yourself against other people.
SLEEP HYGIENE	Habits that help you get good-quality sleep.

QUALITY NOT JUST QUANTITY

- Active use (creating, chatting with friends) vs passive scrolling
- Comparing yourself with edited, filtered posts
- Screens before bed delay sleep
- Notifications break concentration

TAKE CONTROL

- Screen time limits and app timers
- Turn off non-essential notifications; use focus modes
- Charge your phone outside your bedroom
- Train your feed: "not interested", unfollow, mute, reset

ACTIVITY · 25 MINUTES · INDIVIDUAL

MY TECH BALANCE PLAN

1. Estimate your daily screen time by activity (or check your phone's screen time tool at home).
2. Sort your activities into active, passive and comparison-heavy.
3. Do the "how do I feel?" rating for three apps or activities.
4. Write a personal plan with three changes, including at least one feed-control setting.

In your workbook, answer:

? Which activities leave you feeling better or worse?

? Write your three changes.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. What is digital wellbeing?

A A fitness app
B A brand of phone
C Avoiding all technology
D Using technology in a way that supports your health

2. What matters most for wellbeing, according to many researchers?

A Phone brand
B What you do and how it makes you feel, not just total time
C Only total hours
D Wi-Fi speed

3. What is social comparison?

A Judging yourself against others
B A type of feed
C A school report
D Comparing phone prices

4. Which habit helps sleep?

A Watching videos until you fall asleep
B Scrolling in bed
C Notifications on all night
D Charging your phone outside your bedroom

QUICK CHECK: ANSWERS

1. **D: Using technology in a way that supports your health**
   Healthy tech use.

2. **B: What you do and how it makes you feel, not just total time**
   Quality matters.

3. **A: Judging yourself against others**
   Can lower self-esteem.

4. **D: Charging your phone outside your bedroom**
   Better sleep hygiene.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three ways teenagers can take control of their technology use.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe positive and negative effects of technology on wellbeing
- ✓ I can explain why what and how you use tech matters more than total time
- ✓ I can use tools and habits to stay in control of my feeds

LESSON 6 OF 6

DESIGNING FAIRER FEEDS

Governments are starting to regulate algorithms.

STARTER · 5 MINUTES

DO NOW

If you were in charge of a video app's algorithm, what would you want it to optimise for?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how laws and design can make recommendation algorithms fairer and safer, and design an ethical feed

SUCCESS CRITERIA: I CAN...

- ✓ I can describe laws that regulate recommendation algorithms
- ✓ I can explain transparency, user control and what algorithms optimise for
- ✓ I can design an ethical feed and justify my choices

RULES FOR ALGORITHMS

- UK Online Safety Act: platforms must protect children from harmful content
- Ofcom's codes: children's feeds must filter out the most harmful content
- EU Digital Services Act: big platforms must explain recommenders and offer a non-profiled option
- Some countries are setting minimum ages for social media

KEY WORDS

ONLINE SAFETY ACT	A UK law requiring online platforms to reduce the risk of harm to users, especially children.
ALGORITHMIC TRANSPARENCY	Explaining how an algorithm works and why it made a decision.
USER CONTROL	Giving people real choices over how a system works for them.
OPTIMISATION GOAL	What an algorithm is designed to maximise.

WHAT SHOULD A FEED OPTIMISE FOR?

- Engagement alone can reward outrage and extreme content
- Alternatives: satisfaction, learning, connection, wellbeing
- Ask users: "Was this worth your time?"
- Balance relevance with variety and quality

TRANSPARENCY AND CONTROL

- Algorithmic transparency: explain why you're seeing something
- User controls: choose topics, reset, chronological option
- Independent audits of algorithms
- Design with young people, not just for them

ACTIVITY · 35 MINUTES · GROUPS OF 3

FEED REDESIGN CHALLENGE

1. Your team is redesigning the "For You" feed of a (fictional) video app used by 11-16 year olds.

2. Choose your optimisation goal and explain why.

3. Design at least four features for transparency, user control and wellbeing, sketching the screens.

4. Check your design against the Ofcom children's codes and pitch it to the class "board", explaining how the company can still make money.

In your workbook, answer:

? What is your optimisation goal, and why?

? Describe two of your features and how they help.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What does the UK Online Safety Act require platforms to do?

A Delete all algorithms
B Make all apps free
C Reduce the risk of harm, especially to children
D Ban phones

2. What is an optimisation goal?

A What a recommender is built to maximise
B A type of feed
C A password
D A football target

3. What is algorithmic transparency?

A Making screens see-through
B Explaining how an algorithm works and why it made a decision
C Hiding algorithms
D Deleting data

4. What does the EU Digital Services Act require of very large platforms?

A Ban all feeds
B Show only adverts
C Charge users
D Offer a recommender option not based on profiling

QUICK CHECK: ANSWERS

1. **C: Reduce the risk of harm, especially to children**
Protect users.

2. **A: What a recommender is built to maximise**
e.g. engagement.

3. **B: Explaining how an algorithm works and why it made a decision**
"Why am I seeing this?"

4. **D: Offer a recommender option not based on profiling**
User choice.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why optimising a feed only for engagement can cause problems.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe laws that regulate recommendation algorithms
- ✓ I can explain transparency, user control and what algorithms optimise for
- ✓ I can design an ethical feed and justify my choices

END OF TOPIC

ALGORITHMS, FEEDS AND DIGITAL WELLBEING

Revise each lesson in your workbook: the notes, key words and quick checks. Then take the topic test when Mr Butler sets it.